

COINTEGRATION

It is a concept that describes the long run equilibrium relationship between multiple non-stationary time series (the individual comp. of a multivariate time series).

CORE IDEA

1. A TS. Z_t is called integrated of order d on $I(d)$ if the d^{th} diff. $(1-B)^d Z_t$ is stationary. A stationary one is $I(0)$, and random walk $I(1)$.
2. A vector of k time series is said to be cointegrated if individual components are non-stationary but there exists a lin. combin. $W_t = \beta' Z_t$ which is stationary.
3. β is cointegrating vector.
4. If there is m such vectors (lin. indep.) it means the system has m cointegrating relationships and $k-m$ sources of non-stationarity.

EX. VARMA (1,1) model

$$Z_t - \underbrace{\begin{pmatrix} 0.5 & -1.0 \\ -0.25 & 0.5 \end{pmatrix}}_{\text{one of}} Z_{t-1} = a_t - \begin{pmatrix} 0.2 & -0.4 \\ -0.1 & 0.2 \end{pmatrix} a_{t-1}$$

- Nonstationary because the eigenvalues of AR coefficient matrix is 1 (on the unit circle)
- The two eigen vectors of the matrix $\frac{(1.0, -2.0)}{1}, \frac{(0.5, 1.0)}{0}$
- $Y_t = L Z_t$; $L = \begin{pmatrix} 1.0 & -2.0 \\ 0.5 & 1.0 \end{pmatrix}$
 $Y_{1t} = 1.0 Z_{1t} - 2.0 Z_{2t}$
 $Y_{2t} = 0.5 Z_{1t} + 1.0 Z_{2t}$
- $Y_{1t} \Rightarrow$ common trend $I(1)$ Non-stationary
 $Y_{2t} \Rightarrow$ white noise, stationary $I(0)$
- The system is cointegrated, $(0.5, -1.0)'$ is the cointegrating vec.

JUSTIFICATION

- PPP (purchasing power parity) = prices in two countries might wander nonstationarity on their own but theory suggests their relative value should be stable in the long run, which means they are cointegrated.
- Pairs trading = Two stocks in the same industry like company A, and company B is highly likely to be cointegrated. While they fluctuate independently a temporary divergence from their historical ratio is expected to be correct which can be exploited.

VECM (Vector Error correction Model)

Need (PROBLEM)

From B.5.1 we can see that if a system Z_t is cointegrated, just taking the first difference ΔZ_t can solve the non stationarity problem, however it creates new problem, the VMA part becomes noninvertible, a noninvertible model cannot be explained by a finite order VAR model, estimating requires exact maximum likelihood method which is difficult.

SOLUTION (ECM)

To avoid noninvertibility, Engle & Grangers suggested the ECM form. This is a different algebraic representation of the same VARMA model that is stationary and invertible.

ECM form of a VARMA(p, q) model:

$$\Delta Z_t = \Pi Z_{t-1} + \sum_{i=1}^{p-1} \phi_i^* \Delta Z_{t-i} + c(t) + \theta(B)a_t$$

where, i) ΔZ_t (short run change): The stationary differenced series.

ii) $\sum \phi_i^* \Delta Z_{t-i}$ (Short run dynamics): The stationary lagged difference terms that captures the short run relationship of the system.

iii) $c(t) + \theta(B)a_t$ (The errors): This is the stationary and invertible moving average part.

iv) ΠZ_{t-1} (The error correction Term): The matrix $\Pi = -\alpha(1)$ contains all the information about the long run equilibrium.

If the system has m cointegrating vectors rank of Π is m .

$\Pi = \alpha\beta'$, where β is $k \times m$ matrix whose columns are cointegrating vectors.

The term ΠZ_{t-1} is really $\alpha(\beta' Z_{t-1})$. Since $w_{t-1} = \beta' Z_{t-1}$ is the stationary cointegrating relationship (i.e. deviation from long run equilibrium in the previous period) the term shows how the system's short run changes (ΔZ_t) adjust based on the past error. α is the adjustment matrix which decides how the ΔZ_t reacts to previous error.